The examination is being carried out on the **following application documents**

Description, Pages

1-89 as published

Sequence listings, SEQ ID NO

1-4 as published

Claims, Numbers

1-14 filed in electronic form on 03-01-2024

Drawings, Sheets

1/16-16/16 as published

1 *Cited documents*

D1 WANG SHU ET AL: "Intratumoral injection of a CpG oligonucleotide reverts resistance to PD-1 blockade by expanding multifunctional CD8 + T cells", PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES (PNAS), vol. 113, no. 46, 31 October 2016 (2016-10-31), XP093251672, ISSN: 0027-8424, DOI: 10.1073/pnas.1608555113

D2 MARK AYERS ET AL: "IFN-[gamma]-related mRNA profile predicts clinical response to PD-1 blockade", THE JOURNAL OF CLINICAL INVESTIGATION, vol. 127, no. 8, 1 August 2017 (2017-08-01), pages 2930-2940, XP055608325, DOI: 10.1172/JCI91190

D3 HUGO WILLY ET AL: "Genomic and Transcriptomic Features of Response to Anti-PD-1 Therapy in Metastatic Melanoma", CELL, ELSEVIER, AMSTERDAM NL,

vol. 165, no. 1, 17 March 2016 (2016-03-17), pages 35-44, XP029473850, ISSN: 0092-8674, DOI: 10.1016/J.CELL.2016.02.065

2 *Amendments (Art. 123(2) EPC)*

The amendments appear to have a basis in the application as filed and thus comply with Art. 123(2) EPC.

3 *Clarification before search (R. 63(1) EPC)*

3.1 The applicant has been invited to clarify the subject-matter to be searched as the Search Division considered a meaning full search impossible for the following reasons.

3.1.1 Claim 1 refers to *"evaluating a biological sample obtained from the patient for the presence, absence or level of one or more gene associated with a gene ontology (GO) pathway*" followed by a listing of 23 GO pathways. The application does not provide any further explanation or definition of said pathways specified by generic designations, let alone of any of the genes specifically covered by said GO pathways. The nomenclature used for designating said GO pathways does not appear to be common general knowledge either. It is not possible to identify the genes covered by these pathways and it is thus impossible to search for any of the genes covered by the claim as would be necessary to search its whole scope given the "one or more gene" language of claim 1 (emphasis added). Furthermore, claim 1 requires the genes not to be within the said GO pathways, but to be associated therewith, thereby allowing also for even more distantly related genes that are somehow associated with these pathways e.g. by being regulated by and downstream of these pathways. It is not possible to carry out a meaningful search of claim 1. The only instances of genes that are considered to be covered by claim 1 and identifiable by the application are the genes shown in Fig. 2C and 2D, namely CD274 (PDL1), β2m, Tap1, Tap2, Casp1 Gasta3, Rpl41, Rps15, Rps8, Stat1, Stat2 Irf1, Pvr and Ltbr. The search of claim 1 and its dependent claims will thus be limited to these genes.

3.1.2 The foregoing considerations apply *mutatis mutandis* to claim 6 and 7.

3.1.3 Claims 11, 12 and 13 refer to an activation or upregulation of 7 different pathways designated by (gene) names. These pathways do not have a well-recognised meaning in the art and the application lacks a definition of these

pathways, let a lone a limiting definition of when said pathways are considered to be activated or upregulated, except for some exemplifications thereof at p. 59, l. 16 - p. 60, l. 28 which however as such are also not clear. For instance, Trim-family activation may be assayed by modulation of innate immune responses through the control of IFN responsive genes (cf. p. 59, l. 16-21). It is indeterminable what kind of modulation is supposed to occur and which genes are covered by IFN responsive genes and how they are controlled. The same applies to the vague definitions of a Mapk8ip1 pathway activation based on "*functional multiprotein complex in different components of the JNK pathway*" or "*levels of and/or sensitivity to programmed cell death by apoptosis*" (p. 60, l. 7-10); or of the activations of the ELK1 and Lrg1 pathways which are each in turn suggested to be analysed by further ill-defined signalling pathways (cf. p. 60, l. 13-20). Ras and Rap1 pathway activation is suggested to be based on an activation of a protein shown in Fig. 8C and 8D, respectively, but the protein names are illegible in these figure. The only pathway where the application provides an indication of subject-matter to be meaningfully searched is the Arg1 pathway i.e. (i) a hydrolysis of arginine to ornithine and urea, (ii) a suppression of tumor infiltrating lymphocytes (TIls), and (iii) levels or synthesis of polyamines (via L-ornithine). The search of the aforesaid claims will thus be limited to these embodiments of a Arg1 pathway activation.

3.2 In reply to the invitation the applicant argued that claim 9 should be searched focussing on TRIM7 and that "*The indication of TRIM7 can be applied to the other pending claims, in particular claims 11-13*". The latter statement has been interpreted that also the search of claims 1, 7 and 11-13 should relate to TRIM7.

4 *Unity of invention (Art. 82 EPC)*

4.1 The search division considers the application to lack unity of invention and identifies the following separate inventions.

1. claims: 1-3, 6, 7, 9-14(completely); 8(partially)

   Methods as defined in claims 1, 9, 11, 13 and 14; a method as defined by claim 8, but limited to the transgenic non-human animal as defined by claims 6 or 7; wherein each of said methods is limited to TRIM7.

2. claims: 4, 5(completely); 8(partially)

   A method as defined by claim 8, but limited to the transgenic non-human animal made using the composition for use of claim 4 or 5.

4.2 The reasons for the lack of unity of invention are as follows.

4.2.1 Common matter

4.2.1.1 Invention 1 relates to the use of TRIM7 expression to identify cancer cells that are resistant to an anti-PD1 therapy (targeting PD-1, PD-L1 and/or PD-L2), to establish a transgenic non-human animal, to test an anti-cancer drug candidate, and to determine a patient's cancer therapy.

4.2.1.2 Invention 2 relates to providing a composition of one ore more cancer cells (suitable for establishing an transgenic non-human animal) resistant to an anti-PD1 therapy.

4.2.1.3 The only common matter is the resistance of cancer cells to an anti-PD1 therapy. Such cells are however well-known in the art. D1 discloses CT26 cancer cells that are non-responsive to anti-PD1 therapy in a mouse model (Fig. 1A).

4.2.1.4 The common matter therefore cannot constitute a single general inventive concept linking inventions 1 and 2.

4.2.2 Remaining features

4.2.2.1 Invention 1 comprises the additional feature of using the gene expression of TRIM7.

4.2.2.2 Invention 2 comprises the additional feature of selecting cells resistant to anti-PD1 therapy.

4.2.2.3 It follows that the features of inventions1-2 which make a technical contribution over the common matter are different.

4.2.2.4 The technical effect of the additional features of invention 1 is a gene expression profile allowing to identify cells resistant to an anti-PD1 therapy and uses of said gene expression signature. The problem solved by these technical features can be regarded as the provision of a gene expression profile allowing to identify cells resistant to an anti-PD1 therapy and uses of said gene expression signature.

4.2.2.5 The technical effect of the additional features of invention 2 are cancer cells rendered resistant to an anti-PD-1 therapy. The problem solved by these technical features can be regarded as the provision of cancer cells rendered resistant to an anti-PD-1 therapy.

4.2.2.6 As the technical problems are different, the additional technical features cannot be considered to be corresponding special technical features

4.2.3 Conclusion

4.2.3.1 .As the claims do not comprise the same or corresponding special technical features, they lack the technical relationship between their subject-matter required by R. 44 EPC and the claims are not so linked as to form a single general inventive concept.

4.2.3.2 The application therefore does not fulfil the requirement for unity of invention within the meaning of Art. 82 EPC.

4.3 The application can only be prosecuted on the basis of an invention that has been searched by the EPO. Upon entry into the examination phase, the applicant should limit the application to only one searched invention, or unitary group of inventions, and excise those parts of the application relating to the other invention(s) (see EPC Guidelines C-III, 3.2.1 and 3.2.2; H-II, 6.1, 6.3 and 6.4.2).

4.4 The subject-matter to be excised may be made the subject of one or more divisional applications according to R. 36 EPC.

5 *Novelty (Art. 84 EPC)*

None of the prior art discloses a differential TRIM7 expression in cancer cells, let alone an association thereof with a responsiveness/resistance to an anti-PD1 therapy. Therefore, claims 1, 8, 9, 11, 13 and 14 are novel as are their respective dependent claims.

6 *Inventive step (art. 56 EPC)*

6.1 D2 and D3 may be considered the closest prior art for claims 1, 8, 9, 11, 13 and 14 as they disclose gene expression signatures indicative of a cancer's responsiveness/resistance to an anti-PD1 therapy. Claim 1 is distinguished from D2 and D3 in that the expression of TRIM7 is used to select an anti-PD1 cancer therapy. The application shows that in murine cancer cells a reduction of the expression of TRIM7 is associated with a resistance to an anti-PD1 therapy acquired in the course of an anti-PD1 therapy. Thus, the technical problem may be seen in the provision of a gene expression signature allowing to identify cancers that developed or not a resistance to an anti-PD1 therapy in the course of said therapy and are still susceptible or not to said treatment. The prior art does not teach any gene expression signatures of cancers cells as such indicative of an acquired, as opposed to a primary resistance, to an anti-PD1 therapy. Therefore, claim 1 is considered inventive.

6.2 The foregoing considerations appear to apply *mutatis mutandis* to claims 6, 8, 9 and 11.

6.3 Claims 13 and 14 refer to medical uses, wherein attaining a therapeutic effect is a functional feature of the claims. There is no evidence on file that for each of the compounds covered by the claims a therapeutic effect is achieved in patients with an altered expression of TRIM7. Hence, the application lacks enablement across the claimed scope and therefore, the underlying problem of providing a cancer therapy is not solved across the claimed scope. These claims are thus not considered inventive.

7 *Disclosure and clarity (Art. 83 and 84 EPC)*

7.1 In reply to the invitation under R. 63(1) EPC the applicant indicated that the search should be limited with respect to TRIM7. It is however indeterminable how this limitation has to be applied to claims not explicitly mentioning TRIM7 such as claims 1-3 and 6-8. But even for the claims that mention TRIM7 such as claims 9-14 it is indeterminable that all the remaining alternatives are still covered if the genes/pathways are compatible with the limitation on TRIM7. Thus, the scope of all claims is unclear even in view of the limitation to TRIM7.

7.2 The application shows that TRIM7 is downregulated in murine colon cancer cells as they become resistant to an anti-PD1 therapy accompanied by an enhanced inducibility of TRIM7 in these cells by IFNγ (cf. e.g. Fig. 4B and 5C)

7.3 Claims 13 and 14 are considered to lack enablement across the claimed scope for the reasons discussed herein above under item 6.3.

8 *Exceptions (Art. 53(c) EPC)*

Claims 13 and 14 refer to a cancer therapy and an anticancer treatment, respectively, implying or at least encompassing methods of treatment in breach of Art. 53(c) EPC. The claims have nevertheless been interpreted to relate to purpose-related products in accordance with Art. 54(5) EPC.

9 *Exceptions (Art. 53(a) / R. 28(1)(d) EPC)*

Claims 6-8 are considered to relate to subject-matter excepted from patentability pursuant to R. 28(1)(d) EPC.

10 Should examination of the present application be requested, the following also deserves attention.

10.1 The wording "incorporated herein by reference" or the like (p. 1, l. 5; p. 72, l. 12-13; p. 73, l. 11; p. 76, l. 17; p. 88, l. 21) has to be deleted (cf. GL. F-III, 8).

10.2 Last para of p. 88 - last para. of p. 89. have to be deleted pursuant to GL. F-IV, 4.4.